

VJ146A2 High p-lit Gate Sensitivity

Does the VJ/Eskom MLR signal survive looser not-up thresholds?

Reliable Electrification Planning SA

2026-06-04

Purpose

This report tests whether the VJ/Eskom validation signal weakens when the daily darkness/not-up gate is pushed above the current `p_lit < 0.40` threshold. The main report validates 0.05, 0.20, and 0.40; this sensitivity adds 0.50, 0.60, 0.70, 0.80, 0.90, and 0.95.

The setup follows the favored specification in `[[4-work/reports/2026-06-03-recommended-vj-eskom-validation-specification.Rmd]]`:

- standard sample: `gate_tier == "pass"`
- annual-IQR excess-lit removal: `annual_q3_plus_1p5_iqr`
- retained standard sample: 267 nights
- regressor: `shed_share_1_2am_primary`
- denominator: `sum(Eskom MLR from 01:00-02:00 SAST) / sum(RSA Contracted Demand from 01:00-02:00 SAST)`
- inference: Newey-West HAC standard errors with 7-day lag

The temporary reconstruction matches the main-report 0.05, 0.20, and 0.40 columns to numerical precision; the largest absolute difference is 7.08e-16.

Main Pattern

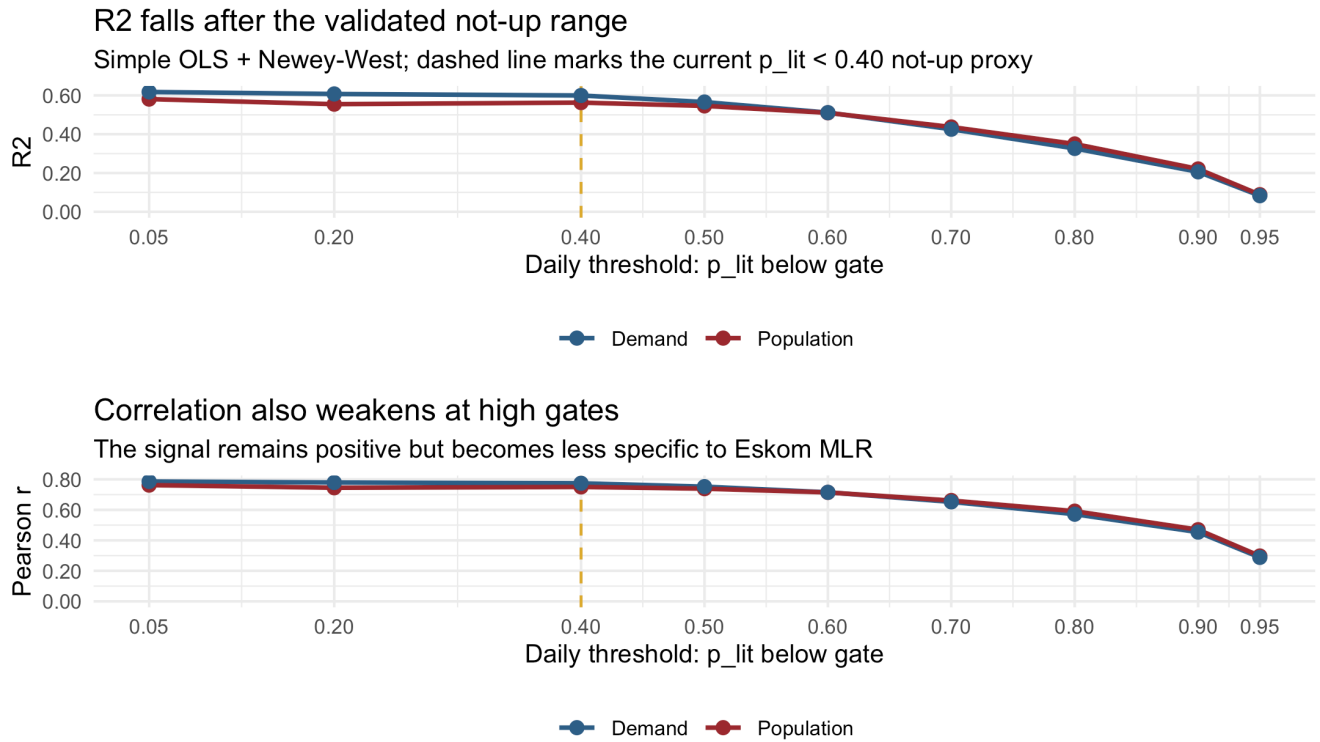


Figure 1: Validation fit degrades as the p_lit gate becomes too loose. R2 is strongest at low-to-moderate thresholds and falls sharply above 0.60, even though coefficient estimates remain positive.

The raw coefficient keeps rising because the outcome becomes broader. A $p_lit < 0.95$ event mechanically includes all lower-threshold dark or not-up events plus many additional partial-lit states. Fit quality is the more informative diagnostic here: in the demand-weighted bridge, R2 is 0.600 at 0.40, but only 0.206 at 0.90 and 0.083 at 0.95.

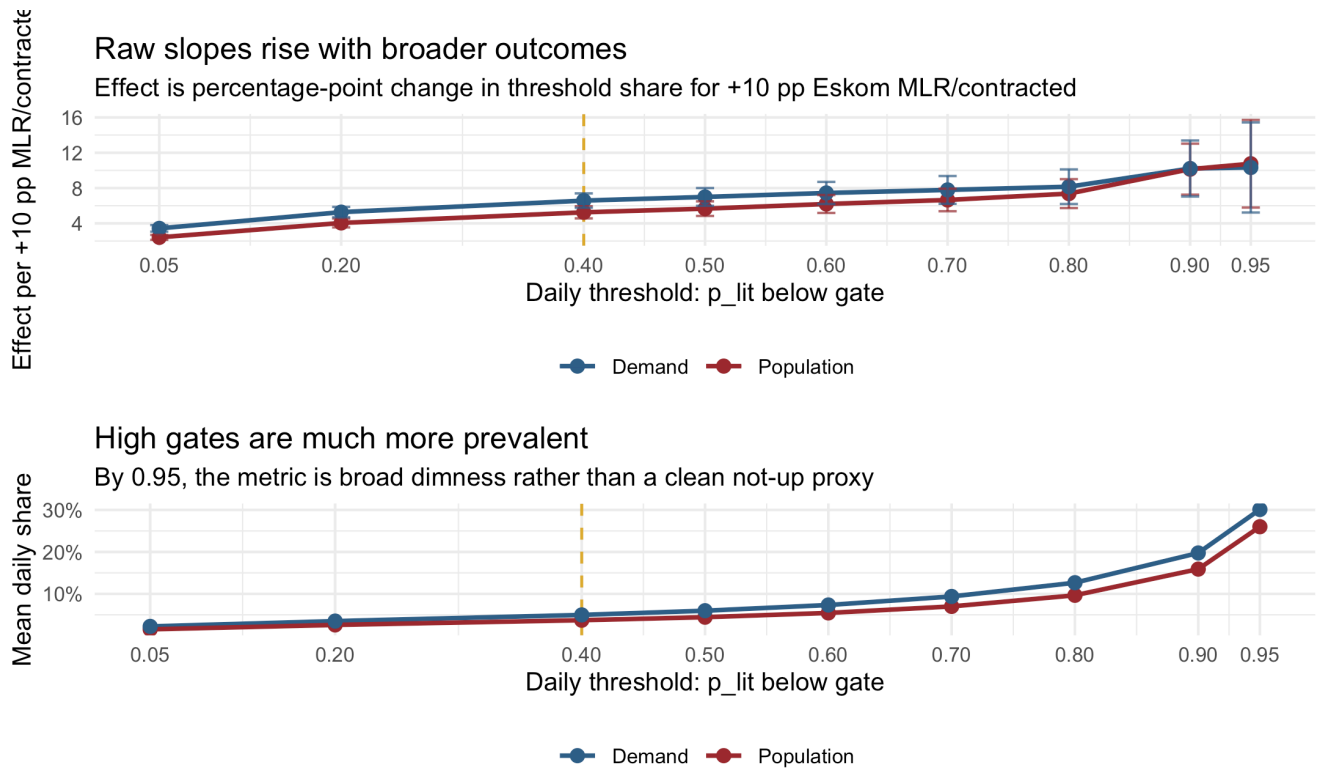


Figure 2: Higher gates have larger raw percentage-point slopes because the outcome is more prevalent, but the increasing slope should not be read as a better validation metric.

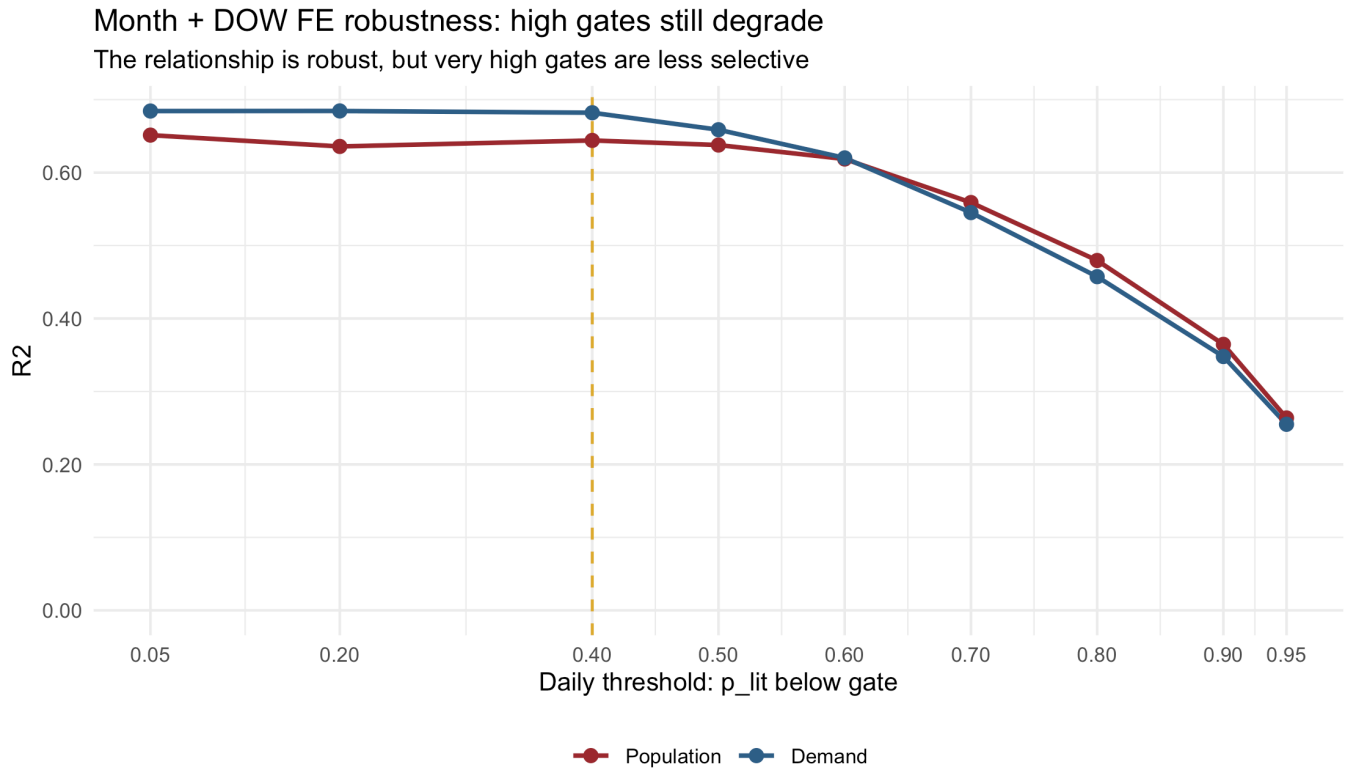


Figure 3: The fixed-effects robustness keeps the same degradation pattern. R2 remains high near 0.40 and falls as the gate approaches 0.95.

Results

Table 1: Simple OLS + Newey-West sensitivity by p-lit gate. The sample and regressor match the favored main-report specification.

Weighting	Gate	Mean outcome	Effect per +10pp MLR	NW SE	NW 95% CI	p-value	Pearson r	R2
Population	0.05	1.55%	2.42 pp	0.14 pp	2.14 to 2.69 pp	7.79e-45	0.762	0.581
Demand	0.05	2.25%	3.43 pp	0.19 pp	3.06 to 3.81 pp	6.87e-48	0.786	0.618
Population	0.20	2.61%	4.05 pp	0.26 pp	3.55 to 4.55 pp	4.75e-40	0.745	0.555
Demand	0.20	3.52%	5.27 pp	0.30 pp	4.69 to 5.86 pp	2.30e-46	0.779	0.608
Population	0.40	3.73%	5.25 pp	0.35 pp	4.56 to 5.94 pp	4.84e-37	0.750	0.563
Demand	0.40	4.99%	6.58 pp	0.42 pp	5.76 to 7.39 pp	3.36e-40	0.774	0.600
Population	0.50	4.45%	5.67 pp	0.42 pp	4.84 to 6.49 pp	8.23e-32	0.739	0.546
Demand	0.50	5.99%	6.99 pp	0.51 pp	5.99 to 8.00 pp	1.84e-32	0.752	0.566
Population	0.60	5.46%	6.19 pp	0.52 pp	5.18 to 7.21 pp	1.09e-26	0.714	0.510
Demand	0.60	7.35%	7.44 pp	0.64 pp	6.19 to 8.69 pp	1.34e-25	0.715	0.511
Population	0.70	7.02%	6.65 pp	0.65 pp	5.38 to 7.93 pp	6.84e-21	0.661	0.437
Demand	0.70	9.38%	7.79 pp	0.80 pp	6.21 to 9.36 pp	3.38e-19	0.652	0.426
Population	0.80	9.66%	7.37 pp	0.84 pp	5.73 to 9.01 pp	1.60e-16	0.591	0.350
Demand	0.80	12.66%	8.15 pp	1.00 pp	6.19 to 10.11 pp	1.41e-14	0.571	0.327
Population	0.90	15.90%	10.14 pp	1.47 pp	7.25 to 13.02 pp	4.22e-11	0.470	0.221
Demand	0.90	19.74%	10.21 pp	1.62 pp	7.04 to 13.38 pp	1.16e-09	0.454	0.206
Population	0.95	26.00%	10.75 pp	2.53 pp	5.79 to 15.71 pp	2.97e-05	0.297	0.088
Demand	0.95	30.11%	10.33 pp	2.60 pp	5.23 to 15.43 pp	9.35e-05	0.288	0.083

Table 2: Month + day-of-week fixed-effects sensitivity by p-lit gate.

Weighting	Gate	Mean outcome	Effect per +10pp MLR	NW SE	NW 95% CI	p-value	R2
Population	0.05	1.55%	2.26 pp	0.17 pp	1.92 to 2.59 pp	1.49e-30	0.651
Demand	0.05	2.25%	3.24 pp	0.22 pp	2.80 to 3.67 pp	4.95e-35	0.684
Population	0.20	2.61%	3.74 pp	0.34 pp	3.07 to 4.41 pp	8.12e-23	0.636
Demand	0.20	3.52%	4.89 pp	0.39 pp	4.14 to 5.65 pp	1.27e-28	0.684
Population	0.40	3.73%	4.76 pp	0.47 pp	3.85 to 5.68 pp	1.15e-20	0.644
Demand	0.40	4.99%	5.98 pp	0.55 pp	4.90 to 7.05 pp	6.51e-23	0.682
Population	0.50	4.45%	5.08 pp	0.53 pp	4.05 to 6.12 pp	8.25e-19	0.638
Demand	0.50	5.99%	6.31 pp	0.65 pp	5.02 to 7.59 pp	7.46e-19	0.659
Population	0.60	5.46%	5.51 pp	0.64 pp	4.25 to 6.76 pp	9.96e-16	0.619
Demand	0.60	7.35%	6.68 pp	0.81 pp	5.10 to 8.26 pp	7.72e-15	0.620
Population	0.70	7.02%	5.83 pp	0.79 pp	4.28 to 7.38 pp	2.54e-12	0.559
Demand	0.70	9.38%	6.94 pp	1.01 pp	4.97 to 8.91 pp	4.17e-11	0.545
Population	0.80	9.66%	6.51 pp	1.02 pp	4.51 to 8.50 pp	8.35e-10	0.480
Demand	0.80	12.66%	7.35 pp	1.25 pp	4.90 to 9.79 pp	1.29e-08	0.457
Population	0.90	15.90%	9.94 pp	1.80 pp	6.40 to 13.48 pp	9.08e-08	0.365
Demand	0.90	19.74%	10.18 pp	1.99 pp	6.27 to 14.08 pp	6.47e-07	0.348
Population	0.95	26.00%	8.33 pp	3.31 pp	1.84 to 14.83 pp	0.0126	0.264
Demand	0.95	30.11%	8.31 pp	3.36 pp	1.72 to 14.89 pp	0.0140	0.255

Settlement-Level Heterogeneity

The aggregate threshold sensitivity can remain strong even when many individual settlements have no detectable response. To check whether the settlement-level picture changes as the gate increases, each settlement was separately regressed on the same 01:00-02:00 Eskom MLR share using the standard validation nights. Settlements with at least 50 observations are retained; constant-outcome settlements are assigned zero slopes, matching the convention used in the main report’s heterogeneity table.

Table 3: Per-settlement slope distribution across p-lit gates. Each settlement is separately regressed on Eskom MLR using standard validation nights with at least 50 observations. Constant-outcome settlements are retained with zero slopes.

Gate	Estimable settlements	Weighted mean slope	Median slope	Weighted median	Population positive	Population sig. positive	Population constant
0.05	5,568	2.75 pp	5.76 pp	0.00 pp	22.4%	14.8%	76.9%
0.20	5,568	4.70 pp	8.59 pp	0.00 pp	29.5%	22.9%	69.8%
0.40	5,568	6.11 pp	10.61 pp	0.00 pp	37.3%	30.9%	62.3%
0.50	5,568	6.62 pp	10.95 pp	0.00 pp	40.1%	33.3%	59.3%
0.60	5,568	7.28 pp	10.92 pp	0.00 pp	48.6%	35.2%	50.8%
0.70	5,568	7.97 pp	10.46 pp	0.99 pp	50.7%	38.1%	48.5%
0.80	5,568	8.93 pp	9.39 pp	4.05 pp	54.8%	43.7%	44.4%
0.90	5,568	11.30 pp	7.06 pp	6.81 pp	81.7%	58.6%	16.3%
0.95	5,568	12.04 pp	5.38 pp	9.15 pp	83.3%	40.8%	12.6%

Per-settlement response changes smoothly with the threshold

Dashed line marks the preferred p-lit < 0.40 not-up proxy

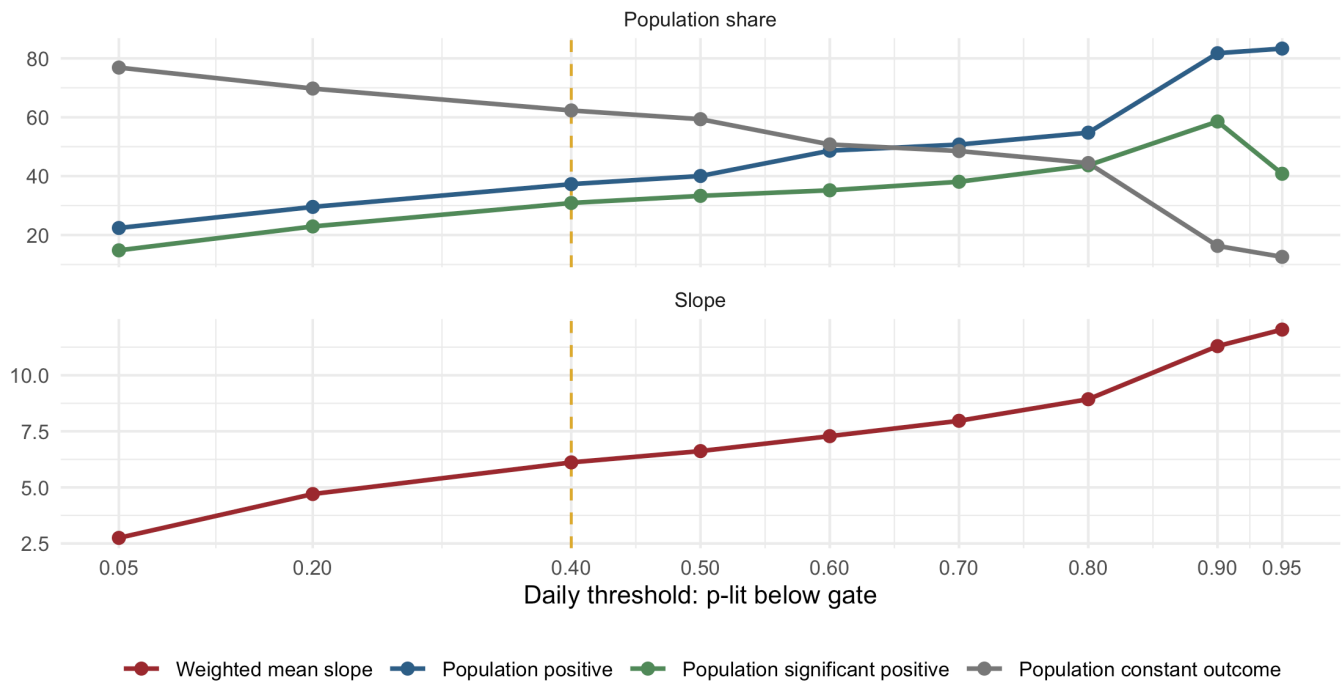


Figure 4: Settlement-level response broadens as the p-lit gate rises. The population-weighted mean slope increases through moderate gates, while the population share with positive slopes rises and the constant-outcome mass falls.

The zero-response mass shrinks as the gate becomes looser

Higher gates pick up broader dimming responses, but they also become less specific in the aggregate validation

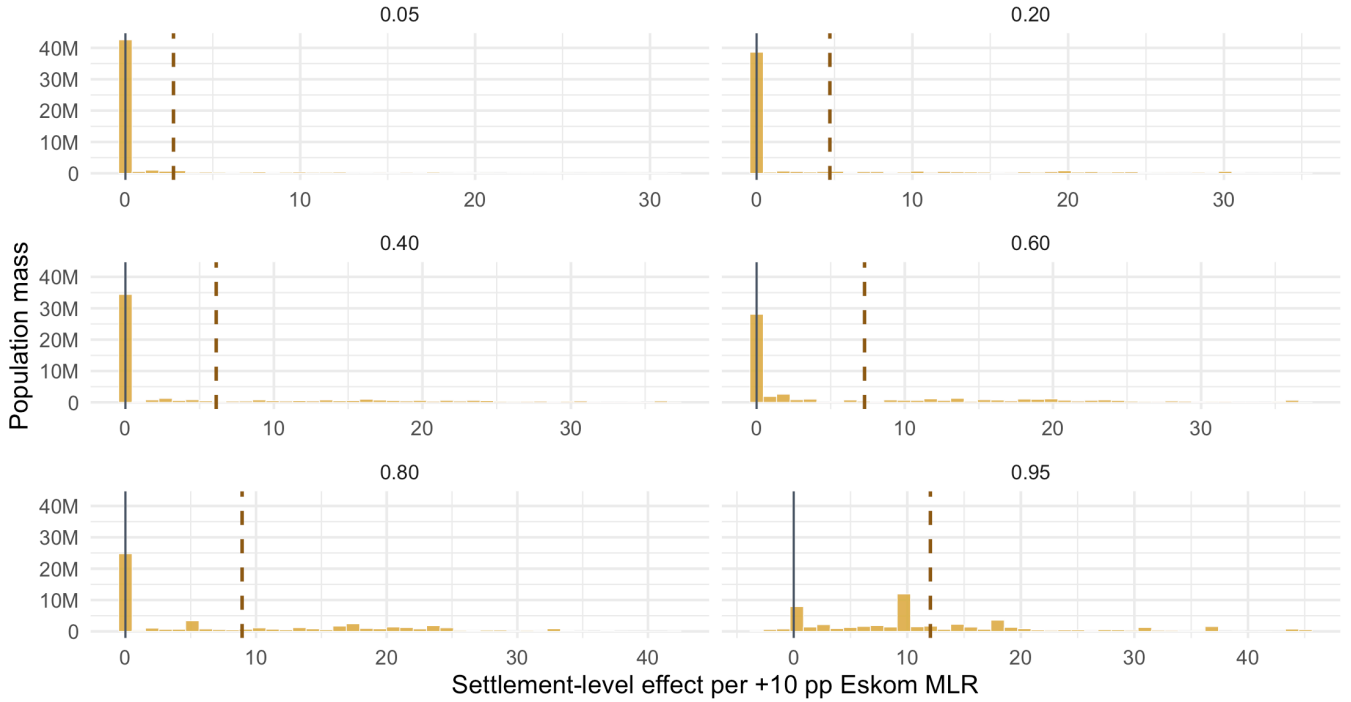


Figure 5: Population-weighted settlement-level slope distributions for selected gates. Histograms are trimmed to each gate's weighted 1st-99th percentile range; dashed lines show the population-weighted mean slope.

The settlement-level graph does change with the gate. At low gates, a large population mass has zero slopes because many settlements are almost never classified strict or mostly dark. As the threshold rises, more settlements enter the response margin and the population-weighted mean slope increases. This helps explain why the aggregate validation remains stable through 0.50 and 0.60: load reduction is moving many otherwise bright settlements into partial-not-up states, not only into full darkness.

That does not mean the highest gates are better. The settlement-level response broadens, but the aggregate validation fit still deteriorates at very high gates. The high-gate distributions show more settlements responding, while the R2 plots above show that those responses are less specifically tied to Eskom MLR once the threshold approaches 0.90 or 0.95.

Does This Undermine the Metric?

No. It supports the interpretation of $p_{lit} < 0.40$ as a practical broad not-up threshold.

The high gates remain statistically significant, but the validation fit deteriorates strongly as the gate rises. That is exactly what we should expect if high gates increasingly include ordinary dim-but-serviceable or partially lit states. In the demand-weighted simple bridge, R2 falls from 0.600 at 0.40 to 0.327 at 0.80, 0.206 at 0.90, and 0.083 at 0.95.

Therefore:

- $p_{lit} < 0.05$ remains the cleanest blackout-like signal.
- $p_{lit} < 0.20$ remains the mostly-dark sensitivity.
- $p_{lit} < 0.40$ remains defensible as the broad not-up / degraded-service proxy for PyPSA.
- thresholds above 0.50 should not be used as the preferred ENS proxy, because they remain significant but become progressively less specific to load reduction.

The validation signal is not fragile. The sensitivity shows a sensible degradation curve: strong at low-to-moderate gates, still present but increasingly diluted at very high gates.